Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: John Vieira Partnership

Physical address of dairy:

Number and Street	City	County	Zip Code
22075 Lone Tree RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0205-0230-0090-0000

B. OPERATORS

Vieira, Daniel

Operator name: Vieira, Daniel

Telephone no.: (209) 838-3904 Landline, (209) 484-7041 Cellular

Mailing Address Number and Street	City	State	Zip Code
22075 Lone Tree RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

C. OWNERS

Vieira, Daniel

Legal owner name: Vieira, Daniel

Telephone no.: (209) 838-3904 Landline, (209) 484-7041 Cellular

Mailing Address Number and Street	City	State	Zip Code
22075 Lone Tree RD	Escalon	CA	95320

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	0	0	0	0
Average number	0	0	0	0	0	0
Avg live weight (lbs)	0	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1.00 *tons per reporting period*

Total nitrogen from manure: 1.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 0.70 *lbs per reporting period*

Total phosphorus from manure: 1.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Well #1	Ground water
Well #2	Ground water
Well #3	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/01/2023	Liquid commercial fertilizer *UN32*	4,340.00 *gal*			32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	11,589.54	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	11,589.54	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	40	40	2	manure	X205-X230-XX09-XXXX
2	37	37	2	manure	X205-X230-XX09-XXXX
3	32	32	1	none	X205-X230-XX09-XXXX
4	20	20	0	none	X205-X230-XX06-XXXX
5	22	22	0	none	X205-X230-XX07-XXXX
6	28	28	0	none	X205-X230-XX07-XXXX
7	16	16	0	none	X205-X230-XX07-XXXX
8 & 9	58	58	0	none	X205-X240-XX01-XXXX
Totals for areas that were used for application	77	77	4		
Totals for areas that were not used for application	176	176	1		
Land application area totals	253	253	5		

B. CROPS AND HARVESTS

1

Field name: 1

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	592.00 *ton*	Dry-weight		57.8	12,000.00	2,500.00	31,000.00		8.11

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.80	149.89	31.23	387.23	1,013.04

1

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	790.00 *ton*	Dry-weight		64.8	11,700.00	1,410.00	11,100.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	19.75	162.68	19.60	154.33	945.47

2

Field name: 2

11/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	520.00 *ton*	Dry-weight		61.7	9,400.00	4,600.00	38,000.00		8.58

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.05	101.19	49.52	409.09	923.67

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	832.00 *ton*	Dry-weight		65.1	11,800.00	1,600.00	10,200.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	22.00	176.00	33.00	145.20	0.00
Total actual harvest content	22.49	185.21	25.11	160.09	973.13

3

Field name: 3

01/01/2001: Pasture

Crop: Pasture Acres planted: 32 Plant date: 01/01/2001

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/15/2023	320.00 *ton*	Dry-weight		67.4	32,200.00	4,160.00	35,200.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	10.00	120.00	30.00	120.00	0.00
Total actual harvest content	10.00	209.94	27.12	229.50	860.64

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/01/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	103.15	33.62	225.65	0.00	100.00 *ton*
Application event totals		103.15	33.62	225.65	0.00	

1 - 06/17/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	10.62	0.00	0.00	305.98	5,040,000.00 *gal*
Application event totals		10.62	0.00	0.00	305.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	12.74	0.00	0.00	367.17	6,048,000.00 *gal*
Application event totals		12.74	0.00	0.00	367.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	150.00	0.00	0.00	0.00	
Application event totals		150.00	0.00	0.00	0.00	

1 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	6.37	0.00	0.00	183.59	3,024,000.00 *gal*
Application event totals		6.37	0.00	0.00	183.59	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	6.37	0.00	0.00	183.59	3,024,000.00 *gal*
Application event totals		6.37	0.00	0.00	183.59	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	6.37	0.00	0.00	183.59	3,024,000.00 *gal*
Application event totals		6.37	0.00	0.00	183.59	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	6.37	0.00	0.00	183.59	3,024,000.00 *gal*
Application event totals		6.37	0.00	0.00	183.59	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	10.62	0.00	0.00	305.98	5,040,000.00 *gal*
Application event totals		10.62	0.00	0.00	305.98	

2 - 11/18/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 - 11/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	111.52	36.34	243.94	0.00	100.00 *ton*
Application event totals		111.52	36.34	243.94	0.00	

2 - 06/22/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	18.37	0.00	0.00	529.26	8,064,000.00 *gal*
Application event totals		18.37	0.00	0.00	529.26	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	18.37	0.00	0.00	529.26	8,064,000.00 *gal*
Application event totals		18.37	0.00	0.00	529.26	

07/17/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	150.00	0.00	0.00	0.00	
Application event totals		150.00	0.00	0.00	0.00	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	13.78	0.00	0.00	396.94	6,048,000.00 *gal*
Application event totals		13.78	0.00	0.00	396.94	

2 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	11.48	0.00	0.00	330.79	5,040,000.00 *gal*
Application event totals		11.48	0.00	0.00	330.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	11.48	0.00	0.00	330.79	5,040,000.00 *gal*
Application event totals		11.48	0.00	0.00	330.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	13.78	0.00	0.00	396.94	6,048,000.00 *gal*
Application event totals		13.78	0.00	0.00	396.94	

3 - 01/01/2001: Pasture

Field name: 3

Crop: Pasture

Plant date: 01/01/2001

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	7.96	0.00	0.00	229.48	3,024,000.00 *gal*
Application event totals		7.96	0.00	0.00	229.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	7.96	0.00	0.00	229.48	3,024,000.00 *gal*
Application event totals		7.96	0.00	0.00	229.48	

3 - 01/01/2001: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	7.96	0.00	0.00	229.48	3,024,000.00 *gal*
Application event totals		7.96	0.00	0.00	229.48	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	7.96	0.00	0.00	229.48	3,024,000.00 *gal*
Application event totals		7.96	0.00	0.00	229.48	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	10.62	0.00	0.00	305.98	4,032,000.00 *gal*
Application event totals		10.62	0.00	0.00	305.98	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	5.31	0.00	0.00	152.99	2,016,000.00 *gal*
Application event totals		5.31	0.00	0.00	152.99	

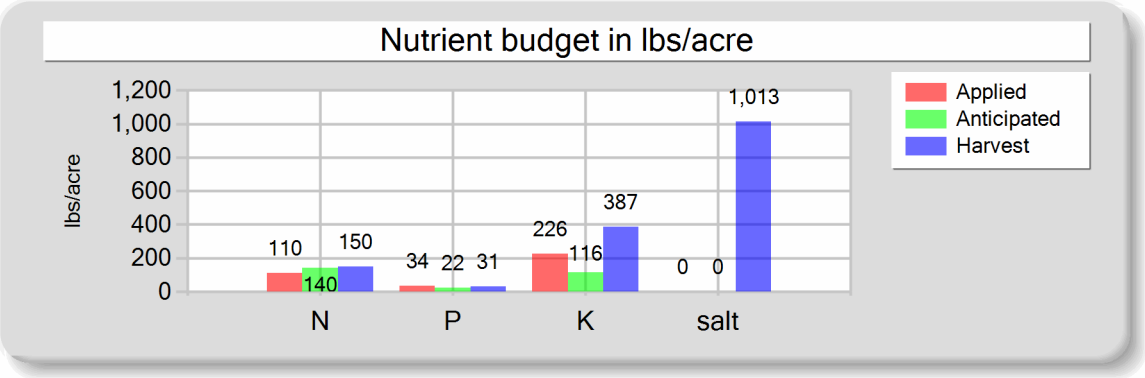
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Well #1	Ground water	7.96	0.00	0.00	229.48	3,024,000.00 *gal*
Application event totals		7.96	0.00	0.00	229.48	

B. NUTRIENT BUDGET

1 - 11/01/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	103.15	33.62	225.65	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	110.15	33.62	225.65	0.00
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	149.89	31.23	387.23	1,013.04
Nutrient balance	-39.74	2.39	-161.58	-1,013.04
Applied to removed ratio	0.73	1.08	0.58	0.00

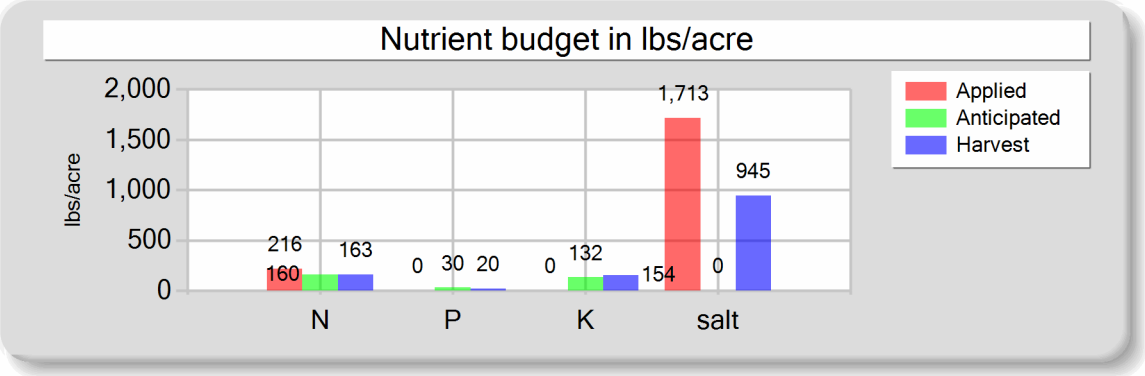
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

1 - 06/17/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	150.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	59.47	0.00	0.00	1,713.48
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	216.47	0.00	0.00	1,713.48
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	162.68	19.60	154.33	945.47
Nutrient balance	53.79	-19.60	-154.33	768.00
Applied to removed ratio	1.33	0.00	0.00	1.81

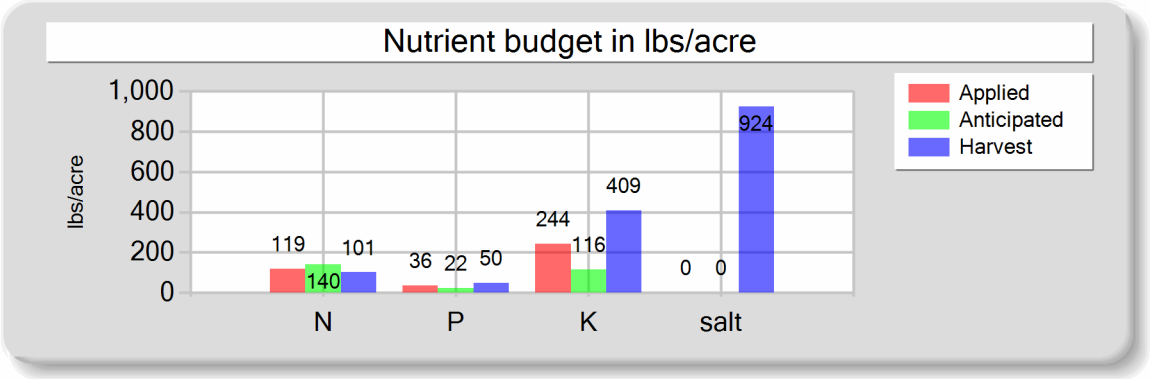
Fresh water applied	
28,224,000.00	*gallons*
1,039.39	*acre-inches*
25.98	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/18/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	111.52	36.34	243.94	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	118.52	36.34	243.94	0.00
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	101.19	49.52	409.09	923.67
Nutrient balance	17.32	-13.18	-165.14	-923.67
Applied to removed ratio	1.17	0.73	0.60	0.00

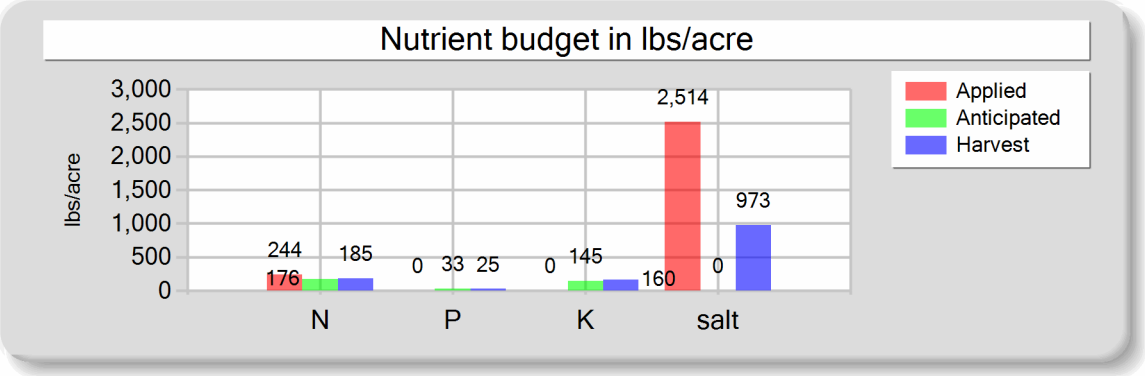
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2 - 06/22/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	150.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	87.25	0.00	0.00	2,513.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	244.25	0.00	0.00	2,513.98
Anticipated crop nutrient removal	176.00	33.00	145.20	0.00
Actual crop nutrient removal	185.21	25.11	160.09	973.13
Nutrient balance	59.05	-25.11	-160.09	1,540.85
Applied to removed ratio	1.32	0.00	0.00	2.58

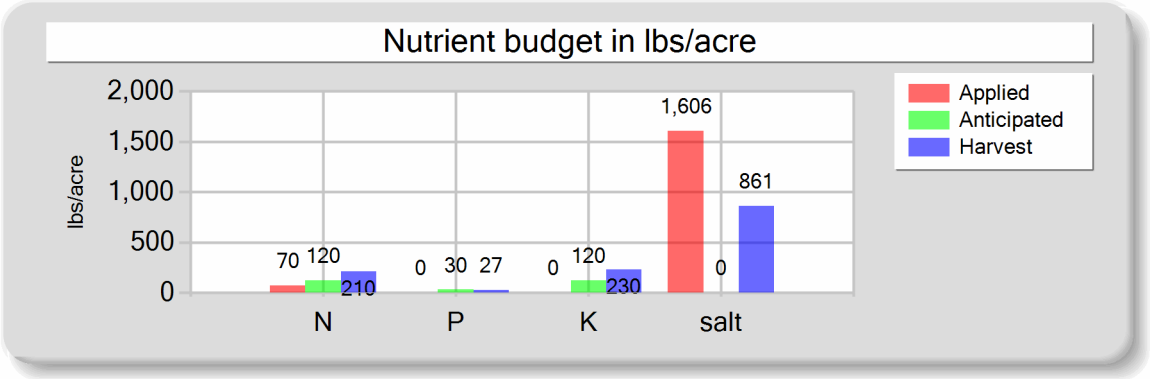
Fresh water applied	
38,304,000.00	*gallons*
1,410.61	*acre-inches*
38.12	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 01/01/2001: Pasture

Field name: 3 Crop: Pasture Plant date: 01/01/2001



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	55.75	0.00	0.00	1,606.38
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	69.75	0.00	0.00	1,606.38
Anticipated crop nutrient removal	120.00	30.00	120.00	0.00
Actual crop nutrient removal	209.94	27.12	229.50	860.64
Nutrient balance	-140.19	-27.12	-229.50	745.74
Applied to removed ratio	0.33	0.00	0.00	1.87

Fresh water applied	
21,168,000.00	*gallons*
779.55	*acre-inches*
24.36	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 05/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	7,300.00	49,000.00							
DL	100.00	10.00	100.00							

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/25/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 25.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	5,630.00	27,200.00							
DL	100.00	10.00	100.00							

B. PROCESS WASTEWATER ANALYSES

No process wastewater analyses entered.

C. FRESH WATER ANALYSES

Well #1

Well #1

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 11/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.10	0.00	10.10								455.00	291
DL	0.10	0.10	0.10								1.00	1

Well #2

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 11/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	16.20	0.00	16.20								895.00	
DL	1.00	0.10	0.10								1.00	

Well #3

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 11/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	39.10	0.00	39.10								1,190.00	
DL	0.10	0.10	0.10								10.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 11/01/2022: Oats, silage-soft dough

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	2,500.00	31,000.00		8.11
DL	100.00	10.00	100.00		0.01

1 - 06/17/2023: Corn, silage

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	1,410.00	11,100.00		6.80
DL	100.00	10.00	100.00		0.01

2 - 11/18/2022: Oats, silage-soft dough

2 - 11/18/2022: Oats, silage-soft dough

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	4,600.00	38,000.00		8.58
DL	10.00	10.00	100.00		0.01

2 - 06/22/2023: Corn, silage

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	1,600.00	10,200.00		6.20
DL	100.00	10.00	100.00		0.01

3 - 01/01/2001: Pasture

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 12/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,200.00	4,160.00	35,200.00		13.20
DL	100.00	10.00	100.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

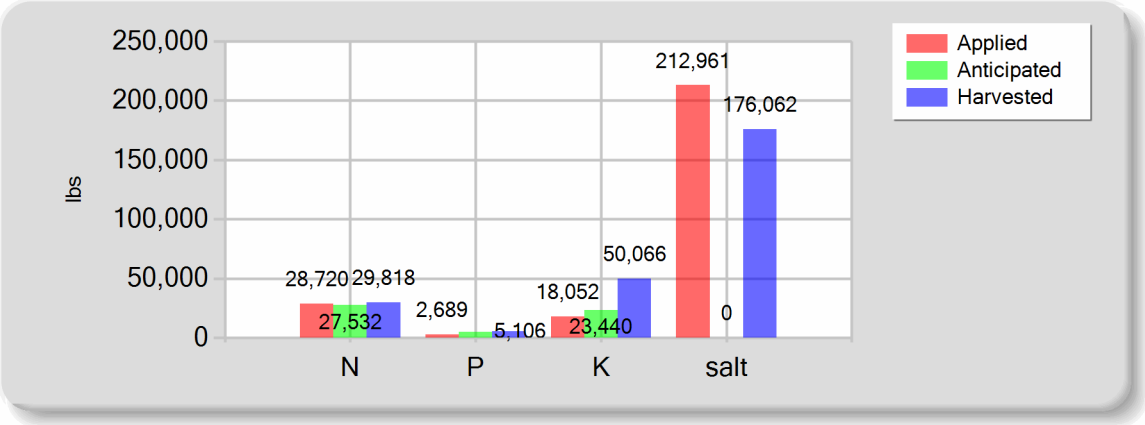
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

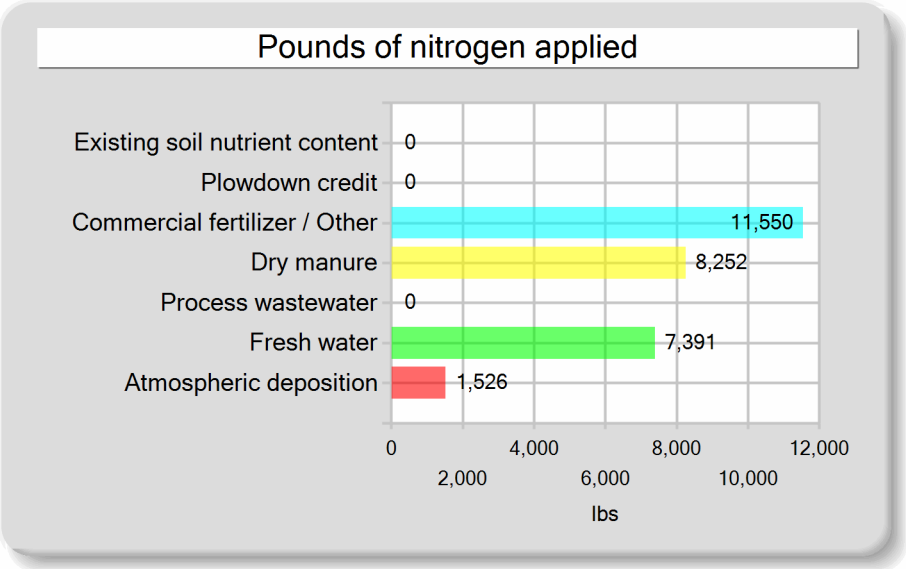
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

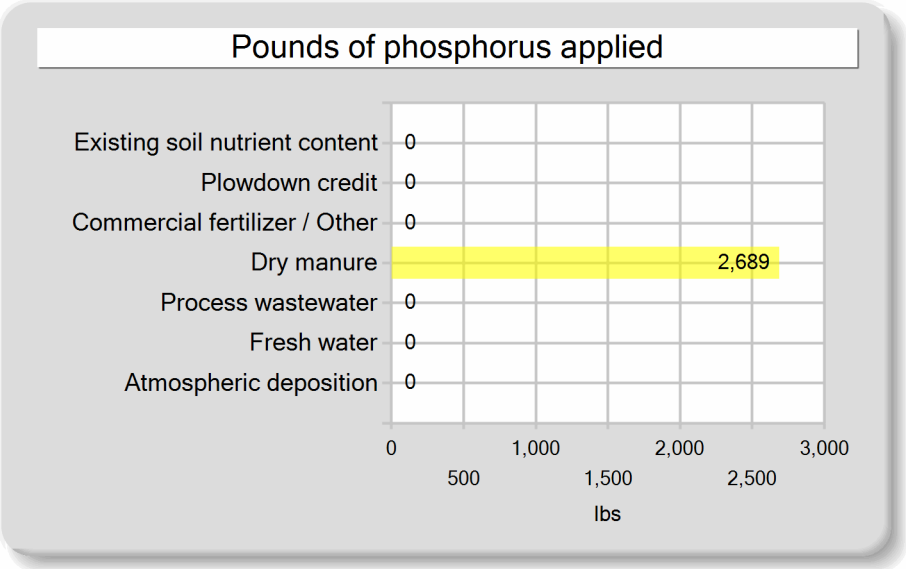
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	11,550.00	0.00	0.00	0.00
Dry manure	8,252.16	2,689.32	18,051.60	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7,391.41	0.00	0.00	212,960.53
Atmospheric deposition	1,526.00	0.00	0.00	0.00
Total nutrients applied	28,719.57	2,689.32	18,051.60	212,960.53
Anticipated crop nutrient removal	27,532.00	5,105.80	23,439.80	0.00
Actual crop nutrient removal	29,817.95	5,662.70	50,066.26	176,062.30
Nutrient balance	-1,098.38	-2,973.38	-32,014.66	36,898.23
Applied to removed ratio	0.96	0.47	0.36	1.21

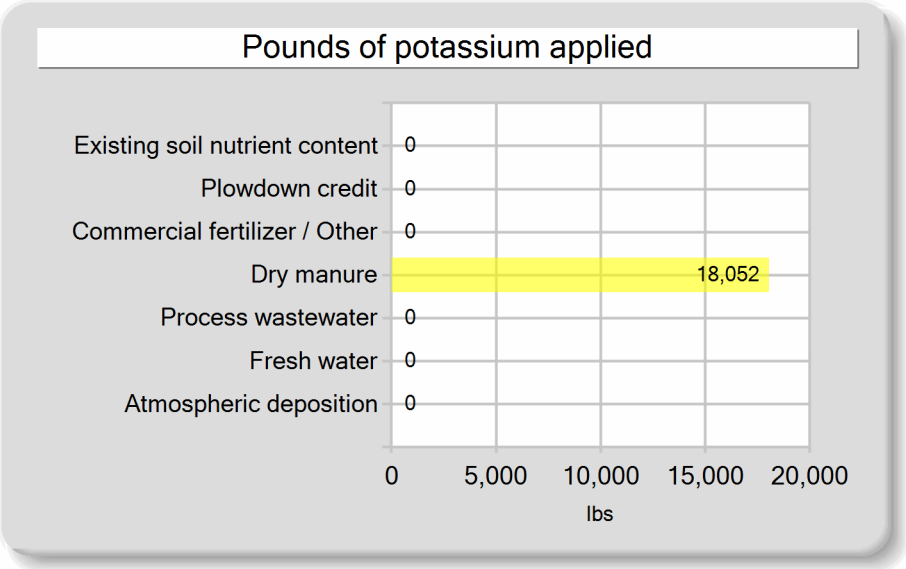
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

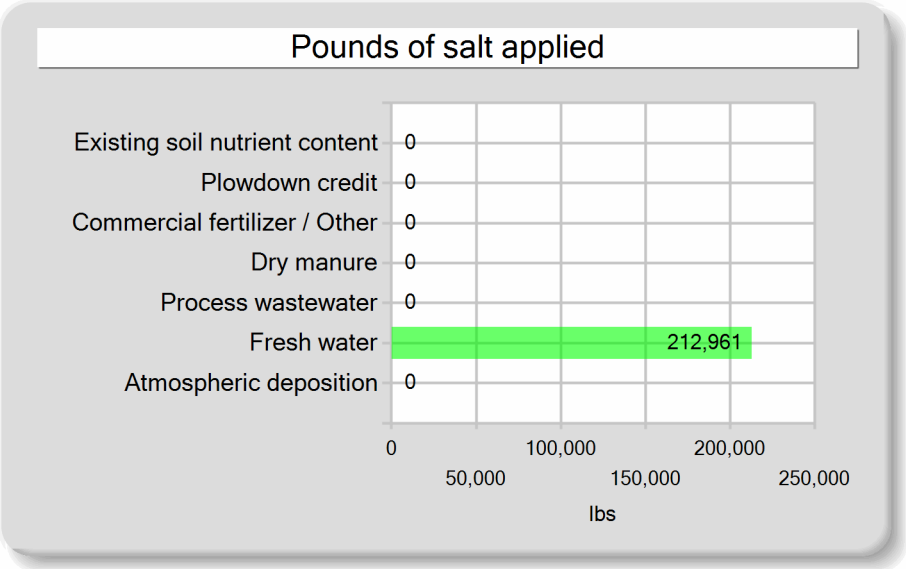


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

The cattle herd has been sold and application areas 4, 5, 6, 7, 8 & 9 have been leased out to a neighbor.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Daniel Vieira	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Dail Vieira	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Daniel Vieira	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-20-24	
DATE	DATE

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
JOHN VIEIRA PARTNERSHIP
22075 E. Lone Tree
Escalon, CA 95320

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER:
RECEIVED DATE: 11/17/23 2:40 PM
SUBMITTED BY: Courier
ANALYZED DATE: 11/17/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well #1	T3321008A	Electrical Conductivity (EC)	EPA 120.1 and	0.455	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	291	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.4			
Electrical Conductivity (EC)	109			0.263
Nitrate Nitrogen (NO3-N)	99.7	98.9 (T3321004A)	101 (T3321004A)	1.88

› warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the ›tomer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714